FIBRES OF THE STRICT HENSELIZATION OF THE NORMALIZATION AND THE MINIMAL PRIMES OVER THE ALGEBRAIC CLOSURE OF THE RESIDUE FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

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Editorial note. The author writes 'small detail omitted' once inside the counted proof span, at the quasi-finiteness step, and states at the start that the proof is 'almost exactly the same as for' the earlier branch lemma before supplying the argument in full; both are reproduced verbatim. The eleven references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Henselization and strict henselization of a local ring, and the residue field of the strict henselization. used at statement lines 31346-31347 and proof lines 31384-31388. Fixes the rings whose spectra are compared and supplies the identification of the residue field with kappa^{sep}.
- The integral closure of a reduced local ring in its total ring of fractions as a finite product of normal domains. used at proof lines 31400-31402. Is what makes every local ring of (A')^{sh} a normal domain and hence gives uniqueness of the minimal prime inside each maximal ideal.
- Henselian pairs, connectedness of V(q) with V of the maximal ideal, and profiniteness of the fibre over the residue field. used at proof lines 31411-31422. Produces the unique maximal ideal containing a given minimal prime, which is the second half of part (3).
- Separability of the residue field extension and the bijection between Hom into the algebraic closure and the spectrum of a tensor product of fields. used at proof lines 31391-31396. Is the identification in part (1) of the fibre with Hom_kappa(kappa', kappa-bar).

Source reference. Original lemma 030C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3943–3943. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 037D: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4196–4196. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00HS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 654–654. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00GU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 620–620. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00GS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 618–618. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 033C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4051–4051. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05WQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7594–7594. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 04MG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6042–6042. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 31155: sources/stacks-more-algebra/pinned-original-source.tex, source lines 31155–31155. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 2753: sources/stacks-more-algebra/pinned-original-source.tex, source lines 2753–2753. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 2948: sources/stacks-more-algebra/pinned-original-source.tex, source lines 2948–2948. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $(A, \mathfrak{m}, \kappa)$ be a local ring. Assume A has finitely many minimal prime ideals. Let A' be the integral closure of A in the total ring of fractions of A_{red}. Choose an algebraic closure $\overline{\kappa}$ of κ and denote $\kappa^{sep} \subset \overline{\kappa}$ the separable algebraic closure of κ. Let A^{sh} be the strict henselization of A with respect to κ^{sep}. Consider the maps*

$$\mathrm{Spec}(A') \xleftarrow{c} \mathrm{Spec}((A')^{sh}) \xrightarrow{e} \mathrm{Spec}(A^{sh})$$

where $(A')^{sh} = A' \otimes_A A^{sh}$. Then

(1) *for $\mathfrak{m}' \subset A'$ maximal the residue field κ' is algebraic over κ and the fibre of c over $\mathfrak{m}'$ can be canonically identified with* $\mathrm{Hom}_\kappa(\kappa', \overline{\kappa})$,
(2) *the right arrow is bijective on minimal primes,*
(3) *every minimal prime of $(A')^{sh}$ is contained in a unique maximal ideal and every maximal ideal contains a unique minimal prime.*

Proof. The proof is almost exactly the same as for Lemma 31155. Let $I \subset A$ be the ideal of nilpotents. We have $(A/I)^{sh} = A^{sh}/IA^{sh}$ by (Algebra, Lemma 05WQ). The spectra of A, A^{sh}, A', and $(A')^h$ are the same as the spectra of A/I, A^{sh}/IA^{sh}, A', and $(A')^{sh} = A' \otimes_{A/I} A^{sh}/IA^{sh}$. Thus we may replace A by $A_{red} = A/I$ and assume A is reduced. Then $A \subset A'$ which we will use below without further mention.

Proof of (1). The field extension κ'/κ is algebraic because A' is integral over A. Since A' is integral over A, we see that $(A')^{sh}$ is integral over A^{sh}. By going up (Algebra, Lemma 00GU) every maximal ideal of A', resp. $(A')^{sh}$ lies over the maximal ideal $\mathfrak{m}$, resp. $\mathfrak{m}^{sh}$ of A, resp. A^h. We have

$$(A')^{sh} \otimes_{A^{sh}} \kappa^{sep} = A' \otimes_A A^h \otimes_{A^h} \kappa^{sep} = (A' \otimes_A \kappa) \otimes_\kappa \kappa^{sep}$$

because the residue field of A^{sh} is κ^{sep}. Thus the fibre of c over $\mathfrak{m}'$ is the spectrum of $\kappa' \otimes_\kappa \kappa^{sep}$. We conclude (1) is true because there is a bijection

$$\mathrm{Hom}_\kappa(\kappa', \overline{\kappa}) \to \mathrm{Spec}(\kappa' \otimes_\kappa \kappa^{sep}), \quad \sigma \mapsto \mathrm{Ker}(\sigma \otimes 1 : \kappa' \otimes_\kappa \kappa^{sep} \to \overline{\kappa})$$

We will use below that the displayed ring is integral over a field hence spectrum of this ring is a profinite space, see Algebra, Lemmas 00GS and 04MG.

Proof of (3). The ring A' is a normal ring and in fact a finite product of normal domains, see Algebra, Lemma 030C. Since A^{sh} is a filtered colimit of étale A-algebras, $(A')^{sh}$ is filtered colimit of étale A'-algebras hence $(A')^{sh}$ is a normal ring by Algebra, Lemmas 033C and 037D. Thus every local ring of $(A')^{sh}$ is a normal domain and we see that every maximal ideal contains a unique minimal prime. By Lemma 2753 applied to $A^{sh} \to (A')^{sh}$ to see that $((A')^{sh}, \mathfrak{m}(A')^{sh})$ is a henselian pair. If $\mathfrak{q} \subset (A')^{sh}$ is a minimal prime (or any prime), then the intersection of $V(\mathfrak{q})$ with $V(\mathfrak{m}(A')^{sh})$ is connected by Lemma 2948 Since $V(\mathfrak{m}(A')^{sh}) = \mathrm{Spec}((A')^{sh} \otimes \kappa^{sh})$ is a profinite space by we see there is a unique maximal ideal containing $\mathfrak{q}$.

Proof of (2). The minimal primes of A' are exactly the primes lying over a minimal prime of A (by construction). Since $A' \to (A')^{sh}$ is flat by going down (Algebra, Lemma 00HS) every minimal prime of $(A')^{sh}$ lies over a minimal prime of A'. Conversely, any prime of $(A')^{sh}$ lying over a minimal prime of A' is minimal because $(A')^{sh}$ is a filtered colimit of étale hence quasi-finite algebras over A' (small detail omitted). We conclude that the minimal primes of $(A')^{sh}$ are exactly the primes which lie over a minimal prime of A. Similarly, the minimal primes of A^{sh} are exactly the primes lying over minimal primes of A. By construction we have $A' \otimes_A Q(A) = Q(A)$ where $Q(A)$ is the total fraction ring of our reduced local ring A. Of course $Q(A)$ is the finite product of residue fields of the minimal primes of A. It follows that

$$(A')^{sh} \otimes_A Q(A) = A^{sh} \otimes_A A' \otimes_A Q(A) = A^{sh} \otimes_A Q(A)$$

Our discussion above shows the spectrum of the ring on the left is the set of minimal primes of $(A')^{sh}$ and the spectrum of the ring on the right is the is the set of minimal primes of A^{sh}. This finishes the proof. □